# RAAHAVI KRISHNAKUMAR


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## SUMMARY


A dedicated Data Science undergraduate at SLIIT with hands-on experience in machine learning, full-stack web development, and mobile application development. Proficient in Python, Java, JavaScript, SQL, and the MERN stack, with practical exposure to ML models including CNN, ARIMA, XGBoost, and Random Forest. Able to work effectively independently and in a team, perform under pressure, and meet deadlines. Seeking an internship or collaborative opportunity to contribute and grow in a data-driven environment.


## EDUCATION

**BSc (Hons) in Information Technology – Specialising in Data Science** — July 2024 – Present
*Sri Lanka Institute of Information Technology (SLIIT)* | *Expected July 2028*

**GCE Advance Level** — 2019
*J/Uduppiddy girls' College*

**GCE Odinary Level** — 2022
*J/Uduppiddy girls' College*

## PROJECTS

### **PharmaAI – Medicine Stock Prediction & Inventory Management System** | Group Project — 2026

*Group Project* | *IT2021 AI & Machine Learning Module (Y2S2)*| *SLIIT*
*github.com/RaahaviKrishanakumar/PharmaAI*

- Built a full-stack MERN app to manage pharmacy inventory, reduce wastage, and enable data-driven stock decisions.
- Led the Sales & Billing module: implemented POS flow, billing logic, invoice functionality, and automatic stock deduction.
- Trained and compared XGBoost, Random Forest, ARIMA, and ANN for demand forecasting; ARIMA selected as best-performing.
- Implemented JWT authentication, bcrypt hashing, low-stock/expiry alerts, and duplicate medicine detection.

**Technologies:** MERNStack (React.js, Node.js, Express.js, MongoDB), XGBoost, Random Forest, ARIMA, ANN, JWT, bcrypt

### **MediCore – Pharmacy Management Mobile App** | Group Project — 2026

*SE2020 Web&Mobile Technology Module (Y2S2)* | *SLIIT* | *github.com/RaahaviKrishanakumar/MediCore*

- Developed a full-stack mobile pharmacy management app covering medicine management, inventory, sales, suppliers, and analytics.
- Contributed the Sales & Billing module: built POS flow, cart handling, checkout, sales records, invoicing, and refund functionality.
- Integrated REST APIs with JWT + role-based access control; deployed backend on Render.

**Technologies:** Expo React Native, Node.js, Express.js, MongoDBAtlas, JWT, REST API, Render

### **Mental Health in Tech Predictor** | Solo Project — 2026

*https://github.com/RaahaviKrishanakumar/Mental-Health-Predictor* |
*Live Demo: https://mental-health-predictor-mg4gnsqdxvhjvww74aywqz.streamlit.app*

- Built an individual ML pipeline to predict mental health treatment-seeking likelihood among tech workers using the OSMI Mental Health in Tech Survey dataset.
- Trained and compared Logistic Regression, Random Forest, and XGBoost; Random Forest selected as the final model based on F1 Score.
- Applied SHAP to identify key workplace risk factors; deployed an interactive Streamlit web app with real-time risk prediction and responsible AI disclaimers.

**Technologies:** Python, Random Forest, Logistic Regression, XGBoost, SHAP, scikit-learn, Streamlit, Joblib, Pandas, NumPy, Matplotlib, Seaborn

### **Paddy Disease Classification System** | Group Project — 2025

*AI & Machine Learning Module (Y2S1)* | *SLIIT Faculty of Computing*

- Applied and evaluated multiple ML models (EfficientNetB0, CNN, MobileNetV2, K-Means) to classify 11 paddy disease categories from image data.
- Best-performing CNN model achieved 94.15% accuracy and a weighted F1-score of 0.9416.
- Compared precision, recall, and F1-score metrics across all models to identify the optimal approach for agricultural disease detection

**Technologies:** Python, CNN, EfficientNetB0, MobileNetV2, K-Means Clustering, scikit-learn, Image Classification

### **Web-Based Voting System for Award Nomination** | Group Project 2025

*Software Engineering Module (Y2S1)*|SLIIT | *github.com/RaahaviKrishanakumar/Web-Based-Voting-System*

- Designed and implemented the Financial Management module for cost tracking and financial data processing within a multi-user secure voting platform.
- Ensured data consistency and accurate calculations across modular system components built with Java Servlets and JSP.

**Technologies:** Java, JSP, Servlets, HTML, CSS, MySQL, Apache Tomcat, GitHub

### **Vehicle Rental System** | Group Project 2025

*Object-Oriented Programming Module (Y1S2)*| *SLIIT* | *github.com/RaahaviKrishanakumar/SLIIT-Vehicle-Rental-System*

- Developed a full-stack web-based vehicle rental management system applying core OOP principles: encapsulation, inheritance, and polymorphism.
- Built front-end with HTML, CSS, and JavaScript; back-end with Java JSP/Servlets.

**Technologies:** Java, JSP, Servlets, HTML, CSS, JavaScript, GitHub

### **Greenhouse Automation System** | Group Project 2024

*Fundamentals of Computing Module (Y1S1)* | *SLIIT*

- Developed a conceptual automation system for monitoring and controlling air temperature and soil moisture in a greenhouse environment.

## CERTIFICATIONS

**AI/ML Engineer–Stage1** 2026
**AI/ML Engineer–Stage2**

*Centre for Open & Distance Education, Faculty of Computing, SLIIT*

**Python for Beginners** 2026
**Python Programming**

*Centre for Open & Distance Learning (CODL), University of Moratuwa*

## SKILLS

**Programming Languages:** Python, Java, JavaScript, SQL
**Web Technologies:** React.js, Node.js, Express.js, HTML, CSS, JSP, Servlets, REST APIs, Apache Tomcat
**Mobile Development:** Expo React Native
**Machine Learning / AI:** CNN, EfficientNetB0, MobileNetV2, K-Means, XGBoost, Random Forest, ARIMA, ANN, scikit-learn, Logistic Regression, Streamlit, SHAP, Joblib
**Data & Visualisation:** SQL, Data Analysis, Confusion Matrix, Classification Reporting
**Databases:** MongoDB, MySQL
**Tools & Platforms:** GitHub, VS Code, IntelliJ IDEA, JWT, bcrypt, Render, Postman, Streamlit
**Soft Skills:** Team Collaboration, Analytical Thinking, Problem Solving, Attention to Detail

## ACHIEVEMENTS & HIGHLIGHTS

- Contributed to a high-accuracy (94.15%) paddy disease classification ML model as part of a 6-member academic research team.
- Successfully delivered six full-stack software projects across consecutive semesters, including MERN stack web apps, a React Native mobile app, and Java web applications.
- Earned AI/ML Engineer Stage 1 & Stage 2 and two Python certifications from SLIIT CODL and University of Moratuwa.
- Independently built and deployed an explainable ML app using SHAP and Streamlit, applying responsible AI principles.
- Open to internships and collaborative projects in Data Science, Machine Learning, and Full-Stack Development.